

# Phase 5 Drone Validation Mission

Generated: 2026-09-12

SCIENTIFIC ACCURACY & INTEGRITY DISCLOSURE:

- **Coordinate System:** LOCAL\_ARBITRARY (Monocular camera optical reference).
- **Scale Status:** RELATIVE\_SCALE (No metric ground coordinates without validated calibration).
- **Georeferencing:** UNREFERENCED (WGS84 / GPS anchors not bound; GeoJSON disabled).
- **Reconstruction:** Authoritative sparse SfM; dense MVS was unexecuted due to hardware constraints and no synthetic dense points were fabricated.

## 1. Mission & Source Video Summary

|                   |                                               |                   |                          |
|-------------------|-----------------------------------------------|-------------------|--------------------------|
| Mission ID        | phase5_drone_validation                       | Operator          | AeroMesh Inspection Team |
| Location / Sector | Operational Flight Zone                       | Mission Type      | Infrastructure           |
| Video Source      | WhatsApp Video 2026-09-01 at 11.27.02 (1).mp4 | Resolution        | 3840x2160                |
| Frame Rate        | 24.0 FPS                                      | Duration / Frames | 30.21 s (725 frames)     |

## 2. AI Detection & Tracking Performance

|                     |                                       |                 |                                  |
|---------------------|---------------------------------------|-----------------|----------------------------------|
| Detector Model      | yolo11n (yolo11n-official)            | Tracker         | Ultralytics persistent ByteTrack |
| Total Detections    | 399                                   | Unique Tracks   | 23                               |
| Confidence Stats    | Mean: 0.495   Min: 0.351   Max: 0.707 | Sampling Rate   | 2.0 FPS (61 frames)              |
| Detections by Class | car: 383, truck: 1, train: 15         | Tracks by Class | car: 21, truck: 1, train: 1      |

## 3. 3D Photogrammetry & Surface Reconstruction

|                      |                                  |                         |                                  |
|----------------------|----------------------------------|-------------------------|----------------------------------|
| SfM Camera Model     | SIMPLE_PINHOLE                   | Registered Cameras      | 20 / 20 (100%)                   |
| Sparse Points        | 12,916                           | Mean Reprojection Error | 0.9785 px                        |
| Surface Mesh Status  | AVAILABLE (pycolmap_poisson)     | Mesh Complexity         | 28,139 vertices   56,120 faces   |
| Dense Reconstruction | UNAVAILABLE (0 synthetic points) | Coordinate Framework    | LOCAL_ARBITRARY / RELATIVE_SCALE |

4. AI-to-3D Multi-View Spatial Fusion

|                         |                                                           |                   |                               |
|-------------------------|-----------------------------------------------------------|-------------------|-------------------------------|
| Authoritative 2D Tracks | 23                                                        | Mean Reproj Error | 2.393 px (threshold: 25.0 px) |
| Tracks Evaluated        | 3                                                         | Acceptance Rate   | 100.0%                        |
| Association Status      | VALID: 1   LOW_CONF: 1  <br>INSUFFICIENT: 1   REJECTED: 0 | Motion States     | STATIC: 3   MOVING: 0         |

| Object ID | Track | Class | Motion | Status                | Position (X, Y, Z)      | Reproj (px) |
|-----------|-------|-------|--------|-----------------------|-------------------------|-------------|
| OBJ_T0001 | T0001 | car   | STATIC | VALID                 | [-17.52, -5.48, 145.64] | 1.95        |
| OBJ_T0002 | T0002 | car   | STATIC | INSUFFICIENT_EVIDENCE | [-13.27, -7.65, 152.11] | N/A         |
| OBJ_T0020 | T0020 | car   | STATIC | LOW_CONFIDENCE        | [-18.48, 0.25, 148.01]  | 8.02        |

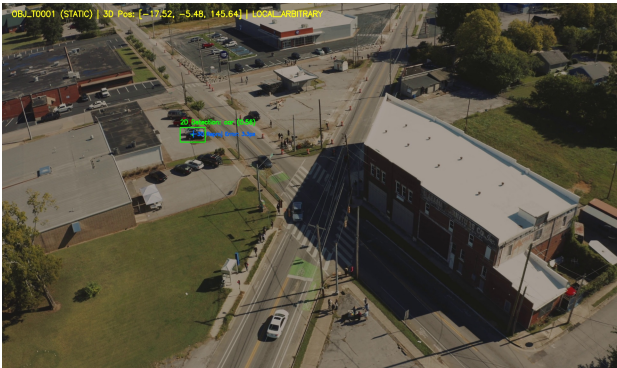
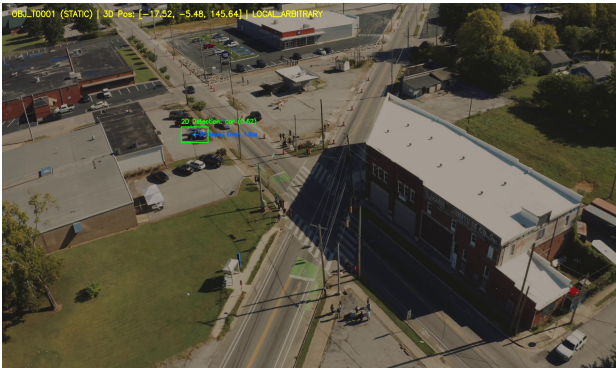
5. Geometric Measurements & Metric Scale Calibration

|                |                                        |                |                           |
|----------------|----------------------------------------|----------------|---------------------------|
| Calibration ID | CAL_phase5_drone_validation_1788531696 | Method         | KNOWN_REFERENCE_DISTANCE  |
| Scale Factor   | 2.39036 m/unit                         | Known Baseline | 15.0 m (Confidence: 0.95) |

| Label / Type                                      | Value                                                              | Unit           | Status                 | Confidence | Uncertainty |
|---------------------------------------------------|--------------------------------------------------------------------|----------------|------------------------|------------|-------------|
| Vehicle Baseline Distance (Calibrated)            | 15.00                                                              | m              | METRIC_CALIBRATED      | 0.95       | ±0.15       |
| Vehicle Baseline Distance (Relative Uncalibrated) | 6.28                                                               | relative_units | RELATIVE               | 0.90       | N/A         |
| Vehicle Dimensions OBJ_T0001 (Calibrated)         | L: 4.54, W: 2.15, H: 1.67                                          | m              | METRIC_CALIBRATED      | 0.85       | ±0.12       |
| Surface Mesh Volume Check                         | Volume computation refused for open surface mesh (not watertight). | m³             | REFUSED_NON_WATERTIGHT | 0.00       | N/A         |

6. Visual Evidence & Reprojection Overlays

The following imagery represents authoritative multi-view observations reprojected onto the 3D model with 2D bounding boxes:



7. Comprehensive Mission Limitations & Disclosures

- **LOCAL\_ARBITRARY:** Reconstruction coordinates are arbitrary relative units, not true meters or GPS coordinates.
- **RELATIVE\_SCALE:** Monocular video SfM is scale-ambiguous; metric distances are strictly valid only where verified ground scale calibration is applied.
- **UNREFERENCED:** Scene is unreferenced against EPSG/WGS84. GeoJSON geographic coordinates (latitude/longitude) are unavailable.
- **DENSE\_MVS\_UNAVAILABLE:** Dense stereo reconstruction requires CUDA or HIP hardware. Sparse SfM is preserved as authoritative geometry without synthetic point fabrication.
- **UNOBSERVED\_SURFACES:** Unobserved, occluded, or low-parallax areas are preserved as unobserved rather than interpolated as synthetic geometry.